SULIT



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Fakulti Komputeran

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION
SEMESTER 2, 2025/2026
— SET C —

SUBJECT CODE	**:**	**SCSM2223**
SUBJECT NAME	**:**	**CROSS-PLATFORM APPLICATION DEVELOPMENT**
SECTION	**:**	**ALL SECTIONS**
TIME	**:**	**(3 HOURS)**
DATE/DAY	**:**	**7 JULY 2026 / TUESDAY**
VENUES	**:**	

INSTRUCTIONS TO THE STUDENTS:

- This exam consists of **THREE (3)** sections with a total of 100 marks: Section A — Multiple-Choice Questions (20 marks), Section B — Structured Questions (45 marks), and Section C — Code Application Questions (35 marks).
- **Answer ALL questions** in the spaces provided in this question booklet.
- You have **THREE (3) HOURS** to complete this exam.
- All code answers must be written using the syntax taught in this course and must be syntactically correct. Comment statements are **NOT** required.
- This is a **CLOSED BOOK** exam. No electronic devices are permitted.

NAME	
MATRIC NO.	
SECTION	
LECTURER	

(This question booklet consists of 17 printed pages including this page)

Mock examination paper (Set C) generated for personal revision practice — not an official UTM examination paper.

SECTION A: MULTIPLE-CHOICE QUESTIONS [20 MARKS]

(INSTRUCTION: This section consists of TWENTY (20) questions. Each question carries ONE (1) mark. Circle the letter of the BEST answer in this booklet.)

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1. HTTP is a STATELESS protocol. This means that:

   A. the server stores a session object for every connected browser
   B. requests can only be sent over an encrypted connection
   C. each request is independent — the server keeps no memory of previous requests
   D. the connection stays open permanently after the first request

2. The tag `<meta name="viewport" content="width=device-width, initial-scale=1.0">` is required so that:

   A. the page is scaled correctly on mobile devices instead of appearing zoomed out
   B. the page automatically loads its CSS file
   C. the browser caches the page for offline use
   D. JavaScript is allowed to run on the page

3. The MAIN advantage of an EXTERNAL stylesheet over inline styles is that:

   A. external CSS supports more properties than inline CSS
   B. inline styles cannot change colours
   C. external CSS loads faster than HTML
   D. one file can style many pages and is cached by the browser, keeping styling consistent and maintainable

4. The difference between `textContent` and `innerHTML` is that:

   A. `textContent` is faster but works only on `<span>` elements
   B. `textContent` inserts the string as plain text, while `innerHTML` parses it as HTML (which can execute injected markup)
   C. `innerHTML` is read-only
   D. they behave identically for all strings

5. The array method `filter()` returns:

   A. a NEW array containing only the elements that pass the test function
   B. the first element that passes the test function
   C. the same array with failing elements set to null
   D. the number of elements that passed the test

6. The result of `JSON.parse('{"id": 7}')` is:

   A. a JSON-formatted string
   B. a syntax error, because JSON keys cannot be numbers
   C. a JavaScript object whose id property equals 7
   D. an array with one element

7. The jQuery statement `$("#msg").text("Saved!")`:

   A. selects all elements with the class `msg`
   B. sets the text content of the element with the ID `msg` to "Saved!"
   C. sends the text "Saved!" to the server
   D. creates a new element with the ID `msg`

8. Consider the program in Figure 1. In which order are the numbers printed?

```
console.log("1");
setTimeout(() => console.log("2"), 0);
console.log("3");
```

Figure 1

   A. 1, 2, 3
   B. 2, 1, 3
   C. 3, 1, 2
   D. 1, 3, 2

9. The `await` keyword may only be used directly inside:

   A. a function declared with the `async` keyword
   B. a `.then()` callback
   C. a `try` block
   D. the global scope of any script

10. In `<input v-model.number="age">`, the `.number` modifier:

   A. restricts the input to a maximum of one digit
   B. rounds the value to the nearest integer
   C. casts the typed value to a Number instead of leaving it a string
   D. prevents the user from typing letters

11. Consider the code in Figure 2. What is the value of `filtered.value`?

```
const items = ref(['Vue', 'Slim', 'PDO'])
const search = ref('s')
const filtered = computed(() =>
  items.value.filter(i => i.toLowerCase().includes(search.value))
)
```

Figure 2

   A. `["Vue"]`
   B. `["Slim"]`
   C. `["Slim", "PDO"]`
   D. `[]`

12. In Vue, a `<slot>` allows:

    A. a child to modify its parent's reactive state directly
    B. two sibling components to share a store
    C. a component to render itself recursively
    D. a parent to inject its own HTML content into a placeholder inside the child's template

13. In Vue Router, defining a route with `component: () => import('../views/TodoView.vue')` means the component is:

    A. loaded lazily — its code is downloaded only when the route is first visited, shrinking the initial bundle
    B. rendered on the server instead of the client
    C. cached forever and never re-rendered
    D. imported twice for reliability

14. An Axios call fails; `err.response` is `undefined` but `err.request` exists. This means:

    A. the server replied with a 4xx status code
    B. the request URL was syntactically invalid
    C. the request was sent but NO response was received — the server is likely down or unreachable
    D. the JSON body could not be parsed

15. A POST to `/students` fails because the `matricNo` already exists (UNIQUE column). The MOST appropriate status code is:

    A. 500 Internal Server Error
    B. 409 Conflict
    C. 204 No Content
    D. 301 Moved Permanently

16. In Slim 4, `$app->addErrorMiddleware(true, true, true)` must be added LAST because:

    A. Slim refuses to start if it is added first
    B. middleware is alphabetically ordered at runtime
    C. error middleware does not support routing
    D. the middleware stack is an onion — the LAST middleware added runs FIRST, so only then can it catch errors thrown by all the other layers

17. A Slim route must read `q` and `limit` from the request `GET /api/books?q=vue&limit=5`. The correct method is:

    A. `$req->getQueryParams()`
    B. `$req->getParsedBody()`
    C. `$args['q']`
    D. `$req->getHeaderLine('q')`

18. Inside a PDO transaction, the second of two UPDATE statements throws an exception. After the catch block runs, the effect of the FIRST update is:

    A. kept permanently — each statement commits itself
    B. undone, because rollBack() cancels every change made since beginTransaction()
    C. kept until the server restarts
    D. converted into an INSERT

19. When `JWT::decode()` is called on a token whose `exp` time has passed, the library:

    A. returns the payload with a warning flag
    B. silently renews the token for another hour
    C. throws an ExpiredException, which the middleware turns into a 401 response
    D. removes the exp claim and continues

20. Which statement about CORS configuration is TRUE?

    A. Access-Control-Allow-Origin can list unlimited origins in one header value
    B. CORS headers are sent by the browser, not the server
    C. pre-flight OPTIONS requests should be answered with 500
    D. Access-Control-Allow-Origin: * CANNOT be combined with Access-Control-Allow-Credentials: true

SECTION B: STRUCTURED QUESTIONS [45 MARKS]

(INSTRUCTION: This section consists of FOUR (4) questions. Answer ALL questions in the spaces provided.)

QUESTION 1 **[8 MARKS]**

Web Architecture and the HTTP Protocol

a) Figure 3 lists the steps of the Slim 4 request lifecycle in the WRONG order. Rewrite the letters in the correct order in which the steps occur.

A. The router matches the URI and dispatches to the route handler
B. The browser sends an HTTP request
C. The middleware stack runs
D. The handler builds the PSR-7 response
E. The web server directs the request to public/index.php
F. Slim parses the request into a PSR-7 request object

Figure 3

(4 marks)

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b) HTTP is stateless, yet the Books API "remembers" who is logged in on every request. Explain how JWT authentication achieves this WITHOUT storing session state on the server.

(2 marks)

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c) Define the two HTTP method properties *safe* and *idempotent*, and classify the GET method against BOTH properties.

(2 marks)

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QUESTION 2 **[10 MARKS]**

Asynchronous JavaScript

a) Consider the program in Figure 4. Assume the network request takes about 200 ms.

```
console.log("start");

setTimeout(() => console.log("timer"), 0);

fetch("https://api.example.com/books")
  .then(() => console.log("data"));

console.log("end");
```

Figure 4

State the EXACT order in which the four words are printed, then justify the position of *"start" and "end"* versus *"timer"* using the JavaScript event loop (call stack and callback queue).

(4 marks)

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b) Write an `async` function `createBook()` that sends the JavaScript object `{ title: "Clean Code", author: "Martin" }` to `POST http://localhost:8000/api/books` using `fetch`. The request must declare the `Content-Type: application/json` header and send the object as a JSON string in the body. After sending, check `response.ok` (throw an `Error` if false), parse the JSON response and log it.

(4 marks)

c) In a Promise chain, state the role of .catch() and .finally().

(2 marks)

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QUESTION 3 **[12 MARKS]**

Vue.js — Reading and Reasoning About Components

a) Figure 5 shows a complete component. Answer the three questions that follow.

```
<template>
  <p>{{ count }} / {{ double }}</p>
  <button @click="count++">+1</button>
</template>

<script setup>
import { ref, computed, watch } from 'vue'

const count = ref(0)
const double = computed(() => count.value * 2)

watch(count, (newV, oldV) => {
  console.log(`count: ${oldV} -> ${newV}`)
})
</script>
```

Figure 5

i) What text does the <p> element show when the page first loads?

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ii) What text does the <p> element show after the button is clicked TWICE?

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iii) Write EVERY line that appears in the console after those two clicks, in order.

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(4 marks)

b) Explain what a <slot> is in Vue and describe how a parent fills a NAMED slot (mention the <template #name> syntax and what happens when the parent provides nothing).

(3 marks)

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c) Figure 6 shows a Pinia store. Answer the three questions that follow.

```
import { defineStore } from 'pinia'
import { ref, computed } from 'vue'

export const useCartStore = defineStore('cart', () => {
  const items = ref([])
  const itemCount = computed(() => items.value.length)
  function add(product) {
    items.value.push(product)
  }
  return { items, itemCount, add }
})
```

Figure 6

i) Identify which member is the state, which is the getter, and which is the action.

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ii) Write the TWO lines a component needs to obtain the store and call `add()` with a product `p`.

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(3 marks)

d) State the purpose of the `v-model` modifiers `.trim` and `.lazy`.

(2 marks)

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QUESTION 4 **[15 MARKS]**

Backend Security and Deployment — Scenario

Your team's *BookHub* application (Vue 3 + Slim 4 + MySQL + JWT) is preparing for public release.

a) Compare SESSION-BASED authentication with TOKEN-BASED (JWT) authentication, and state which one this course uses for its REST API and WHY it fits the REST architecture.

(3 marks)

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b) Figure 7 shows the ownership check inside a protected route. Answer the three questions that follow.

```
$book = findBook((int)$args['id']);
if (!$book) {
    return jsonResponse($res, ['error' => 'not found'], 404);
}

$auth = $req->getAttribute('auth');
if ((int)$book['created_by'] !== (int)$auth['sub']
        && $auth['role'] !== 'admin') {
    return jsonResponse($res, ['error' => 'forbidden'], 403);
}
```

Figure 7

i) Name the attack class this code prevents.

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ii) Explain where the `auth` attribute read by `getAttribute('auth')` comes from.

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iii) Explain why the first branch returns 404 but the second returns 403.

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(4 marks)

c) Define *defense in depth*, then name the FIVE layers taught in this course with ONE example control for EACH layer.

(4 marks)

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d) Figure 8 lists the release steps for the Android version of BookHub in the WRONG order. Rewrite the letters in the correct order, then explain why the step in letter D must happen BEFORE the step in letter A.

```
A. npx cap sync android
B. Set VITE_API_BASE_URL to the production API URL
C. Build the signed AAB in Android Studio
D. npm run build
```

E. npx cap open android

Figure 8

(4 marks)

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SECTION C: CODE APPLICATION QUESTIONS [35 MARKS]

(INSTRUCTION: This section consists of THREE (3) questions. Answer ALL questions. Write syntactically correct code in the spaces provided.)

QUESTION 5 **[10 MARKS]**

DOM Debugging, jQuery and CSS

a) The program in Figure 9 should add the typed task to the list when the button is clicked, but it contains **THREE (3) errors**. Identify each error and rewrite the corrected program.

```
const btn = document.getElementByld("addBtn");

btn.addEventListener(click, function () {
  const input = document.getElementById("task");
  const item = document.createElement("li");
  item.textContent = input.value;
  document.querySelector(".taskList").appendChild(item);
  input.value = "";
});

<!-- HTML: -->
<input id="task">
<button id="addBtn">Add</button>
<ul id="taskList"></ul>
```

Figure 9

(4 marks)

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b) Using jQuery, write code that runs only after the DOM is ready, and on clicking `#loadBtn` performs `$.get` on `"data.json"`, then writes the `message` property of the returned data into the element `#out` using `.text()`.

(3 marks)

c) Write CSS that: declares a custom property `--brand` with the value `#42B883` on `:root`; uses it for the background colour of `.btn`; and darkens `.btn` to `#2f8e63` on hover.

(3 marks)

QUESTION 6 **[10 MARKS]**

Vue Component — Filtered List

Write a COMPLETE Vue 3 Single File Component named `BookSearch.vue` using `<script setup>` that satisfies ALL of the following requirements:

i) receives a prop named `books` (Array, required) where each book is an object `{ id, title }`;

ii) declares a reactive string `searchText` bound to a text input with `v-model`;

iii) declares a `computed` property `filteredBooks` containing only the books whose `title` contains `searchText`, case-insensitive;

iv) renders `filteredBooks` as list items (`v-for` with `:key` on the book's `id`) showing each title;

v) shows the paragraph `<p>No books found.</p>` using `v-if` when `filteredBooks` is empty.

(10 marks)

QUESTION 7 **[15 MARKS]**

Full-Stack Backend: Slim 4, PDO Transactions and JWT Middleware

a) Using a Slim 4 **route group** with the prefix `/api`, write the route `DELETE /api/books/{id}` inside the group. The route must: cast the route parameter to an integer; call the provided function `deleteBook($id)` which returns `true` if a row was deleted, `false` otherwise; return **200** with `{"message":"Book deleted"}` on success, or **404** with `{"error":"Book not found"}` on failure. Every response must carry `Content-Type: application/json`.

(5 marks)

b) A purchase must update TWO tables atomically: decrease the stock in `books` and insert a row into `orders`. Write PHP using a PDO **TRANSACTION** that: begins the transaction; executes `UPDATE books SET stock = stock - 1 WHERE id = :id` and `INSERT INTO orders (book_id, user_id) VALUES (:bid, :uid)` as prepared statements (values in `$bookId`, `$userId`); commits on success; and on ANY `PDOException` rolls back and re-throws the exception.

(5 marks)

c) Complete the core of a JWT middleware: read the `Authorization` header from the request, remove the `Bearer` prefix, then decode the token inside `try/catch` using `JWT::decode()` with `new Key($secret, 'HS256')`. On failure, the catch block must produce a **401** JSON response (assume `jsonResponse($res, $data, $status)` exists).

(3 marks)

d) The `POST /auth/login` endpoint is being brute-forced. Name the defence taught in this course, state the HTTP status code it returns when the limit is exceeded, and name the response header that tells the client when to retry.

(2 marks)

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END OF QUESTION PAPER